

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

Ph 2227
(Basic Quantum Mechanics and Solid State Physics)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) What is the necessity of quantum theory to explain photoelectric effect? The work function of different types of metal are different. Is it true? Argue your answer. (10)
(b) Derive an expression for change in wavelength for a photon that is scattered through an angle ϕ by a particle of rest mass m_0 in the Compton effect. Show the condition for maximum wavelength change. (15)
(c) The threshold wavelength for tungsten is 2300 Å. What will be the energy of the electron ejected from the surface by light of wavelength 1800 Å? (10)
2. (a) Applying uncertainty principle show that an electron cannot be the part of a nucleus. (07)
(b) According to Bohr model under what condition an electron can radiate electromagnetic energy? Mention the quantum numbers associated with Vector atom model. (18)
(c) If the Rydberg Constant is $1.097 \times 10^7 \text{ m}^{-1}$, then find the wavelength of the least energetic photon in the Balmer spectrum and the wavelength of the series limit for the Lyman Series. (10)
3. (a) What is quantum mechanics? State the Postulates of Quantum mechanics. (10)
(b) The one-dimensional time independent Schrödinger equation is (15)

$$\left(-\frac{\hbar^2}{2m}\right)\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

- i. Give the meanings of the symbols in this equation.
- ii. A particle of mass 'm' is contained in a one dimensional box of width 'a'. The potential energy $V(x)$ is infinite at the walls of the box ($x=0$ and $x=a$) and zero in between ($0 < x < a$). Solve the Schrödinger equation for this particle and hence show that the normalized solution have the form

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \text{ with energy}$$

$$E_n = \frac{\hbar^2 n^2}{8ma^2}, \text{ where 'n' is an integer.}$$

- iii. Sketch the wave functions and the Probability density distribution for the cases $n=1, 2$ and 3.

- (c) Explain the meaning of the eigen-functions and eigen values. Show that (10)

$$\frac{d\langle P_x \rangle}{dt} = \langle F_x \rangle.$$

4. (a) If the orbital angular momentum is expressed as $\vec{L} = \vec{r} \times \vec{p}$, then obtain all the angular momentum operators in 3D system. (12)
- (b) What is understood by stationary state and probability current density? Show that the Probability density ρ and Probability current density $\vec{j}$, satisfy the equation of continuity (13)
- $$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0.$$
- (c) Find the first excited state of the Harmonic Oscillator. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is the difference between nanomaterials and nanotechnology? Give the classification of nanomaterials. (12)
- (b) Explain top-down and bottom-up methods for synthesis of nanomaterials. (13)
- (c) Write short notes on quantum dots and give some applications of nanoparticles. (10)
6. (a) What is meant by phonon? Write down the assumptions made by Einstein in calculating lattice heat capacity in solids. (10)
- (b) Following Einstein theory deduce an expression for the lattice heat capacity. Discuss the successes and failures of this model. (15)
- (c) Gold has the same structure as copper. The velocity of sound in gold is 2100 ms^{-1} and that in copper is 3800 ms^{-1} . If the Debye temperature of copper is 348 K , determine the Debye temperature of gold. The densities of gold and copper are respectively $1.93 \times 10^4 \text{ kgm}^{-3}$ and 8960 kgm^{-3} and their atomic weights are 197.0 and 63.54 a.m.u. respectively. (10)
7. (a) What is electron gas? Show that the average kinetic energy of an electron at absolute zero is $\overline{E_0} = \frac{3}{5} E_{F_0}$. (10)
- (b) What is Hall effect? Deduce an expression for Hall coefficient for a metal and show the relation between Hall coefficient and Hall mobility. (15)
- (c) The atomic radius of sodium is $1.86 \text{ \AA}$. Calculate the Fermi energy of sodium at absolute zero. (10)
8. (a) Discuss in detail the concept of directionality, monochromaticity, intensity and Coherence of LASER light. (12)
- (b) Define induced absorption and stimulated emission. Describe the working principle of Ruby LASER with suitable diagram. (13)
- (c) The Coherence Length for the red cadmium line of wavelength $6.5 \times 10^{-7} \text{ m}$ is 30 cm . Calculate (10)
- (i) The number of oscillations corresponding to the Coherence Length and
- (ii) The Coherence time.

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

MSE 2201
(Phase Diagrams and Transformations)

Time: 3.0 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Define the following terms: (10)
(i) Component of a system; (ii) Phase; (iii) Grain; (iv) Microstructure; (v) Phase diagram.
(b) Describe how coring happens during solidification of a solid solution alloy under typical industrial conditions using a hypothetical binary equilibrium phase diagram. (15)
(c) Calculate the degrees of freedom for a 50:50 copper-nickel alloy at (i) 1400°C, where it exists as a single liquid phase; (ii) 1300°C, where it exists as a two-phase mixture of liquid and solid solutions; and (iii) 1200°C, where it exists as a single solid-solution phase. Assume a constant pressure of 1 atm in each case. (10)
2. (a) Using Figure: 2(a), in 200g of a lead-tin alloy of eutectic composition, determine (15)
 - i. the amount and composition of each phase immediately after the reaction has been completed
 - ii. calculate the mass of phases present
 - iii. calculate the masses of lead and tin in each phase.

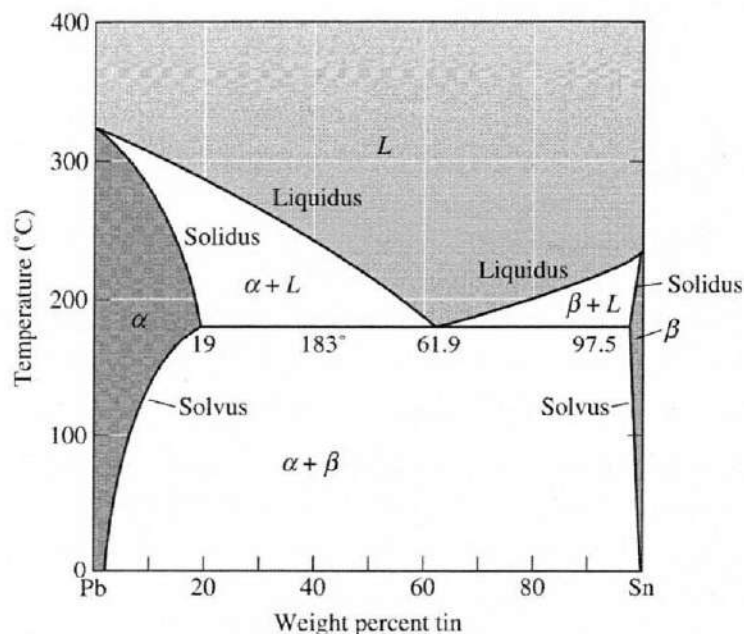


Figure: 2(a)

- (b) Define stoichiometric and nonstoichiometric intermetallic compounds. (05)
- (c) Explain the mechanism of irregular eutectic growth during solidification with necessary sketches. (15)
3. (a) Draw a complete Iron-Iron Carbide diagram showing different phases at different temperatures and compositions. (15)
(b) Discuss the changes that take place during the very slow cooling of a hypereutectic cast-iron containing 4.3 percent carbon. (15)
(c) Draw the cooling curve of pure iron. (05)

4. (a) Take three compositions R, S and L as shown in Figure: 4(a) and mix two parts of R with three parts of S and five parts of L. Calculate the composition of the mixture. (10)

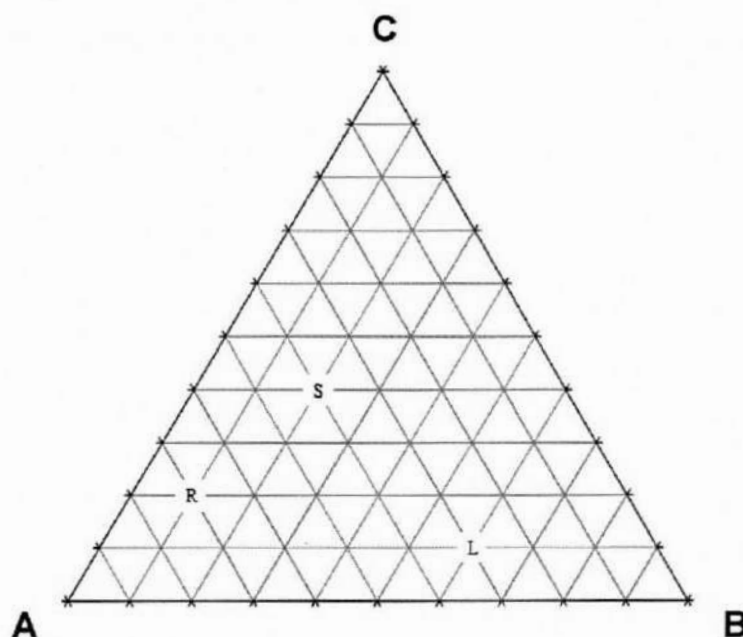


Figure: 4(a)

- (b) Describe the microstructural changes that occur in a low carbon steel containing 0.3% carbon during slow cooling from austenite range to room temperature. (15)
- (c) For a 99.65 wt% Fe - 0.35 wt% C alloy at a temperature just below the eutectoid, determine the followings: (10)
- the fractions of total ferrite and cementite phases
 - the fractions of the pro-eutectoid ferrite and pearlite
 - the fractions of eutectoid ferrite.

Section B

(Answer ANY THREE questions from this section in **answer script B**)

5. (a) What is the effect of temperature on diffusion? Derive the equation for substitutional diffusion. (15)
- (b) Define homogeneous and heterogeneous nucleation. "Activation energy barrier (ΔG_{het}^*) for heterogeneous nucleation is smaller than homogeneous nucleation but critical radius r^* is same for both nucleation", explain in details with schematic diagrams. (20)
6. (a) Define nucleation and growth in solidification with schematic diagram. How does shape factor $S(\theta)$ govern the formation of heterogeneous nucleation? (15)
- (b) What is meant by undercooling? How does the degree of undercooling affect the critical size of the nucleus? (10)
- (c) In γ and α iron which interstitial position is mostly favorable for carbon atom diffusion and why? Explain with lattice diagram. (10)
7. (a) How does austenite phase of FCC crystal structure transform to martensite phase? Explain with proper schematic diagram. (15)
- (b) Define diffusionless transformation. Explain the characteristics of diffusionless transformation. (10)
- (c) "Martensite formation is an athermal process", justify. (10)
8. (a) How does retained austenite affect the martensite phase and how it can be removed? (10)
- (b) What is meant by Gibbs' Equilateral Triangle? Plot a general phase diagram of a ternary alloy system and describe by Gibbs' Phase rule. (15)
- (c) Define coherent, semicoherent and incoherent interfaces. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

MSE 2203
(Mechanical Behavior of Materials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Draw a neat stress-strain diagram of ductile materials and show how you would measure the toughness and modulus of resilience from the graph. (08)
(b) Two points on a stress-strain curve for a material are $\sigma=278$ MPa at $\epsilon=0.08$ and $\sigma=322$ MPa at $\epsilon=0.16$ respectively. Find the strain hardening exponent and strength coefficient of the material and predict σ at $\epsilon=0.20$. [Assume Power Law for calculations] (15)
(c) What is strain hardening? Explain the causes of strain hardening in ductile materials with sketch. (12)
2. (a) Briefly discuss the restoring steps of ductility after work hardening with figures. (15)
(b) Discuss the strain fields surrounding an edge dislocation and their interaction with each other using proper sketches. (10)
(c) The density of BCC iron is 7.882 g/cc and the lattice parameter is 0.2866 nm when H_2 atoms are introduced at interstitial positions. Calculate (i) the atomic fraction of H_2 atoms per unit cell and (ii) the required number of unit cell on average that contains one H_2 atom. (10)
3. (a) How do point defects in ionic structures differ those found in pure elements? Explain with proper figures. (10)
(b) Derive an equation of Schmid Law ($\tau = \alpha M$) for a tensile sample with proper sketch. (15)
(c) What is meant by slip system? Show the number of slip system in $\{110\}$, $\{211\}$ and $\{321\}$ slip planes of a BCC structure using figures. (10)
4. (a) Why doesn't slip occur uniformly at an atomic scale? Explain it with the concept of Frank Read source. (10)
(b) 'Climb is a non-conservative motion of dislocation', explain the reasons. (06)
(c) What is Bauschinger effect? Explain the causes of Bauschinger effect. (09)
(d) Draw the Mohr's circle of the stress element shown in figure 4(d) using graph paper. Determine the principle stresses and maximum shear stresses from the graph. (10)

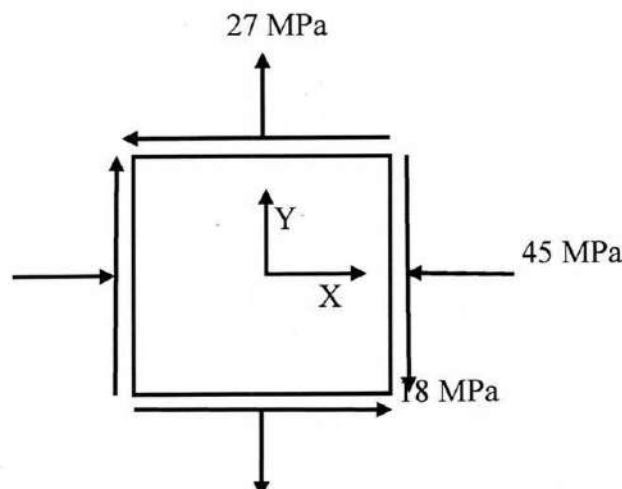


Figure: 4(d)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Briefly explain the crack-tip opening displacement (CTOD) and crack-opening displacement (COD) theories. (10)
- (b) State and explain the Griffith theory and show that the fracture stress is inversely proportional to the square root of the crack length. (15)
- (c) Determine the theoretical fracture strength of a silica fiber. If the modulus of elasticity is 95 GPa, surface-creation energy is 1.5 J/m^2 and equilibrium atomic distance of two silica atoms is 0.15 nm. Also, calculate the practical fracture strength of the same material in the presence of a 150 nm length crack. (10)
6. (a) Draw Goodman diagram and explain the dependency of limiting range of stress on mean stress. (15)
- (b) "A notch in a thick plate is far more damaging than in a thin plate"- Justify the statement with the necessary figures. (10)
- (c) The stress intensity for a partial-through thickness flaw is given by $K = \sigma\sqrt{\pi a} \sqrt{\sec \frac{\pi a}{2t}}$ where 'a' is the depth of penetration of the flaw through a wall thickness t. If the flaw is 5mm deep in a wall 12mm thick, determine whether the wall will support a stress of 172 MPa if it is made from 7075-T6 aluminium alloy. Fracture toughness of 7075-T6 aluminium alloy is $24 \text{ MPa}\sqrt{\text{m}}$. (10)
7. (a) What is fatigue fracture? Explain the effect of completely non-reversed cyclic stress on fatigue using a Goodman diagram. (15)
- (b) Explain the DBTT curve formation process from Charpy impact test and also mention the significance of this curve. (10)
- (c) For the cyclic stress-strain curve, $\Delta\sigma = 150 \text{ MPa}$ and $\Delta\varepsilon = 0.00129$. If $\varepsilon_f = 0.30$ and $E = 22 \times 10^4 \text{ MPa}$, determine (i) $\Delta\varepsilon_e$ and $\Delta\varepsilon_p$; (ii) the number of cycles to failure. [Use Coffin-Manson relation for calculation] (10)
8. (a) What is creep failure? Explain the creep process with the aid of a creep curve. (15)
- (b) As a materials engineer, what are the properties of an alloy you should consider while selecting for high temperature application, explain them briefly. (10)
- (c) Differentiate between creep curve and stress rupture curve using appropriate diagram. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

CSE 2227
(Programming Techniques)

Time: 3.0 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Give a brief description of type conversion and type casting with examples. (11)
(b) Write a program in C to print the output of the following series without using any iteration statement. (12)
$$1^2 + 2^2 + 3^2 + \dots + n^2$$

(c) Write a function in C to check whether a number is palindrome or not. (12)
2. (a) Write a function to find the second maximum number and second minimum number in an one-dimensional array. (12)
(b) Write a C program to find the transpose of a square matrix. (12)
(c) Describe entry-controlled loop and exit-controlled loop with proper flow-chart and examples. (11)
3. (a) Write a C function to update a square logic matrix with central horizontal and vertical values 1. The number of rows or columns (n) is always an odd number. For example, when n=5, the matrix will be like shown in fig. 3(a). You can't use nested for-loop, you have to use only one for-loop. (10)

0	0	1	0	0
0	0	1	0	0
1	1	1	1	1
0	0	1	0	0
0	0	1	0	0

Fig. 3(a)

- (b) Write a C program to solve the equation $ax^2 + bx + c = 0$. (10)
- (c) Write short notes on the following: (i) Integer constants; (ii) Real constants; (iii) Single character constants; (iv) Static variable; (v) Global variable. (15)
4. (a) Write a C program to print a pyramid pattern. For example, when n=5, the output will be like shown in fig. 4(a). (12)

					*					
				*	*	*				
			*	*	*	*	*			
		*	*	*	*	*	*	*		
*	*	*	*	*	*	*	*	*	*	

Fig. 4(a)

- (b) Write a C program to search a specific number and print its index in a two-dimensional array. (12)
- (c) Why should goto be avoided in C? Differentiate between call by value and call by reference with examples. (11)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Distinguish between the following terms: (i) Objects and Classes; (ii) Accessors and Mutations. (08)
- (b) What is inline function? Why do we need inline function in C++? (07)
- (c) What is a friend function? Write down the characteristics of a friend function. (07)
- (d) Material scientists work with diverse types of materials that have different melting points. We are going to design a class namely 'Material' to store the name and melting point of various materials. Now, (13)
- i. Design a parameterized constructor whose role is to initialize the name and melting point of different materials.
- ii. Create a function that calculates and displays the average melting point of two materials when objects of the 'Material' class are passed to it.
6. (a) Explain about the following built-in string functions of C with example. (12)
- (i) Strcpy(); (ii) Strcat(); (iii) Strlen(); (iv) Strcmp().
- (b) Write a program in C to find a sub string from a given string. (14)
- (c) Write a program in C to check if a given string is palindrome or not. (09)
7. (a) Write short notes of the followings: (i) Dereferencing operators; (ii) Address of operator; (12)
- (iii) Null pointer; (iv) Void pointer.
- (b) Write a program using pointers to read an array of integers and print it in reverse order. (10)
- (c) Differentiate between malloc() and calloc(). (06)
- (d) Consider the following program in Fig. 7(d). Is there any problem in the program? If there is, (07)
- what is the problem and how can you solve this?

<pre>#include < stdio.h > void func (); int main () { int i; for (i = 0; i < 10000; i + +) { func (); } return 0; }</pre>	<pre>void func () { int *ptr; int i; ptr = (int*) malloc(256*sizeof(int)); for (i = 0; i < 256; i + +) { scanf("%d", &ptr[i]); } }</pre>
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Fig. 7(d)

8. (a) What is a union? How does a union differ from a structure? Explain with example. (06)
- (b) Define a structure data type called 'time_struct' containing three members integer hour, integer minute, and integer second. Write a program in C to add two times using the structure data type 'time_struct' and display the output in the following form: (12)
- hour : minute : second
- (c) Write a program to copy one file to another. While doing so, replace all lowercase characters to their equivalent uppercase characters. (10)
- (d) What is the significance of EOF? Why do we need to close a file? (07)

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B.Sc. Engineering 2nd Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

Math 2227
(Statistics, Fourier Series and Numerical Analysis)

Time: 3.0 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Evaluate the followings (the interval of differencing being unity) (10)

(i) $\Delta^2 \left(\frac{1}{x^2+x-2} \right)$ and (ii) $\left(\frac{\Delta^2}{E} \right) x^3$.

- (b) Estimate the missing term from the following table: (10)

x	10	15	20	25	30	35
$f(x)$	43	-	29	32	-	77

- (c) The number of candidates who secured marks between certain limits in an examination as follows. Estimate the number of candidates getting marks less than 70. (15)

Number of candidates	Marks
41	0-19
62	20-39
65	40-59
50	60-79
17	80-99

2. (a) Establish the relation between simple difference and divided difference by taking four unknown components. (09)

- (b) The distances (x , cm) traversed by a particle at different times (t , seconds) are given below: (13)

t	0.0	0.1	0.2	0.3	0.4
x	3.01	3.16	3.29	3.36	3.40

- (c) Use Stirling formula to find y_{35} , given $y_{20} = 512$, $y_{30} = 439$, $y_{40} = 346$, $y_{50} = 243$. (13)

3. (a) The population of a town is given below in difference time. Find the rate of growth of the population in 1921. (13)

Year	1921	1931	1941	1951	1961
Population (in thousand)	19.96	38.65	58.81	77.21	94.61

- (b) Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using (i) Simpson's $1/3$ rule and (ii) Weddle's rule by taking 7 equidistant ordinates. (12)

- (c) Write the iteration schem of Newton-Raphson method of $f(x) = 0$ and discuss the advantage and disadvantage of the method. (10)

4. (a) Find a real root of the equation $2x = \log_{10} x + 7$ by using the Fixed point iteration method, correct up to four decimal places. (12)

- (b) Solve the following system of equations by Gauss-Jordan elimination method: (11)

$$2x + 3y - z = 5, \quad 4x + 4y - 3z = 3, \quad -2x + 3y - z = 1.$$

- (c) Solve the following system of linear equation by Gauss-Jacobi method (upto second iteration): * (12)
- $$x + 17y - 2z = 48; \quad 2x + 2y + 18z = 30; \quad 30x - 2y + 3z = 48.$$

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) The mean and Standard deviation of scores obtained in Mathematics of 60 students of MSE-2K19 batch were found to be 58.25 and 6.3 respectively. Later on it was found on scrutiny that the scores of 56 and 64 of two students has been wrongly recorded as 65 and 46 respectively. What is the corrected mean and Standard deviation corresponding to the corrected figures? (10)

- (b) The frequency distribution are given below: (17)

Class	40-50	50-60	60-70	70-80
Frequency	10	15	20	5

Calculate the first four moments about (origin) 'O' from the above table. Also find skewness and comment on the result.

- (c) The diameter of an electric cable is assumed to be a continuous random variable with *p.d.f.* (08)
- $$f(x) = 6x(1 - x); \quad 0 \leq x \leq 1.$$

6. (a) The probability that a machine will produce all bolts in a production run within specification is 0.97. A sample of 8 machines is taken at random. Calculate the probability that (i) all 8; (ii) 7 or 8; (iii) at least 6 machines will produce all bolts within specification. (12)

- (b) The chances of solving a problem in statistics of *A* and *B* are $\frac{1}{6}$ and $\frac{1}{8}$ respectively. If the probability of their making a common error is $\frac{1}{525}$ and they obtain the same result. Find the probability that their answer is correct. (11)

- (c) In a sample of 1000 cases, the mean of a certain test is 14 and variance is 6.25. Assuming the distribution to be normal. Find (i) how many student score between 12 and 15? and (ii) how many score above 18? [Use the attached Z table] (12)

7. (a) Find the simple linear regression of number of Heavy Hydrogen atom on energy escaped from an object. The experimental data are given below: (15)

Atom	3	5	2
Energy	5	8	3

- (b) Find the co-efficient of co-relation between temperature and strength of bonding of a solid material. The experimental data are given below: (13)

Temperature	15	20	30	40
Strength	7	11	20	32

- (c) Write down the characteristics of normal probability curve. (07)

8. (a) Write down the conditions for existence of Fourier series. (05)

- (b) Draw the graph of the function, (12)

$$f(x) = 0, -\pi < x < 0$$

$$= x^2, 0 < x < \pi$$

- (c) The following table connects values of *x* and *y* for a statistical input: (18)

<i>x</i>	0	1	2	3	4	5
<i>y</i>	9	28	24	28	26	20

Express *y* in Fourier series upto second harmonic.